

Ridge Broker Ablation Experiment

Question and design. These 3 ablations restrict the broker’s Ridge learning setup. They are not an additive decomposition of its advantage. The base Ridge reference pools mediated-match observations and uses additive and bilinear pair features. *Size-matched pooled history* retains those features but limits the broker to the smaller of its available sample and the median agent fitting sample. *Single-principal history* randomly retains one endpoint type from each mediated match. At prediction, its fitted one-type function scores both parties, with the fitting-sample output mean subtracted once. *Additive Ridge* retains both endpoint types but replaces the pair features with their type-vector sum. The latter two variants cannot represent bilinear interactions. Agent parameters remain fixed, although matching can change agents’ realized histories. All conditions use $\lambda_a = 0.001$ and $\lambda_b = 0.001$, the same fitting schedule, and the same broker observation cap before size matching.

The primary estimand in realization c is the late-period broker-minus-principal difference in holdout rank correlation

$$G_v(c) = \bar{r}_{b,v}(c) - \bar{r}_{a,v}(c),$$

for Ridge broker variant v . The ablation contrast is $G_v(c) - G_{\text{pair}}(c)$. Each bar averages periods 481–500 within seed and then averages seeds. The baseline comparisons use 50 common seeds. Other conditions use 20 common seeds. The $\rho \times \delta$ analysis gives one observation to each of the 31 effective realizations, including the baseline. Repeated grid coordinates that map to the same realization receive no additional weight. Because the ablations alter broker rankings, matches, and later learning histories, the contrasts describe the resulting systems, not fixed-data comparisons of fitted regressions.

Baseline. Table RA1 reports late-period baseline outcomes. Figure RA1A shows the full baseline trajectory rather than only its late-period summary.

Table RA1: Baseline outcomes by Ridge broker variant

Outcome	Pair	Size-matched	Single-principal	Additive
Agent holdout rank correlation	0.471	0.721	0.697	0.652
Broker holdout rank correlation	0.845	0.757	0.444	0.474
Broker-minus-principal difference in holdout rank correlation	0.374	0.035	-0.253	-0.179
Brokered-minus-self output gap	1.564	1.264	0.574	0.490
Outsourcing rate	0.942	0.758	0.508	0.574

Entries use all 50 common seeds. Each seed is first averaged over periods 481–500.

Results across effective realizations. Table RA2 summarizes the primary outcome over the baseline and $\rho \times \delta$ design. Figure RA1B–D shows the paired realization contrasts, while Figure RA2 displays the $\rho \times \delta$ pattern for each broker variant.

Table RA2: Broker-minus-principal difference in holdout rank correlation across effective realizations

Variant	Median difference	Median Δ	Δ [P10, P90]	Difference > 0	$\Delta < 0$
Pair	0.440	—	—	31/31	—
Size-matched	0.056	-0.370	[-0.447, -0.221]	27/31	31/31
Single-principal	-0.253	-0.597	[-1.087, -0.275]	8/31	31/31
Additive	-0.203	-0.553	[-1.053, -0.069]	10/31	30/31

Δ is the ablation-minus-pair contrast. Each effective realization receives one observation.

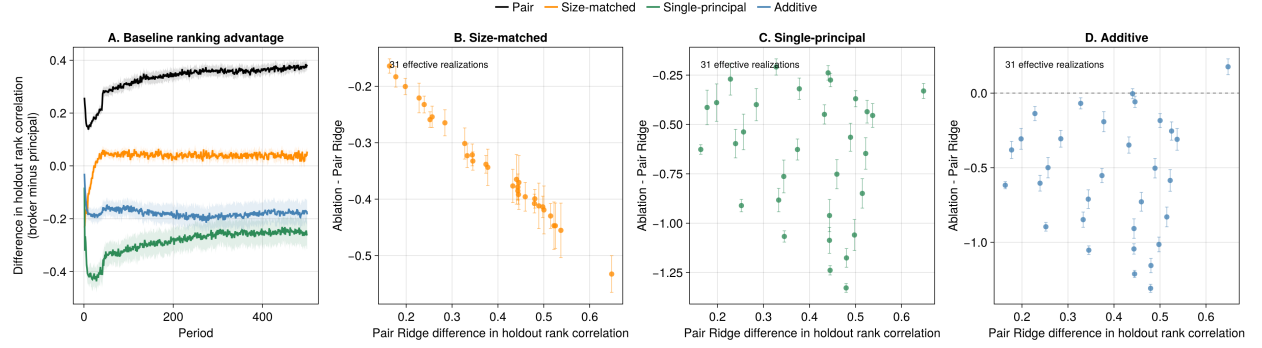


Figure RA1: Ridge broker restrictions. Panel A shows the baseline broker-minus-principal difference in holdout rank correlation over time under base Ridge and the 3 restrictions. Lines are ensemble means over 50 common seeds; ribbons are pointwise 95% Monte Carlo intervals. Panels B–D show each restriction’s paired change from base Ridge across the 31 effective regimes, against that regime’s base Ridge mean. Whiskers are paired 95% Monte Carlo intervals for the mean seed-level change, and dashed lines mark no change.

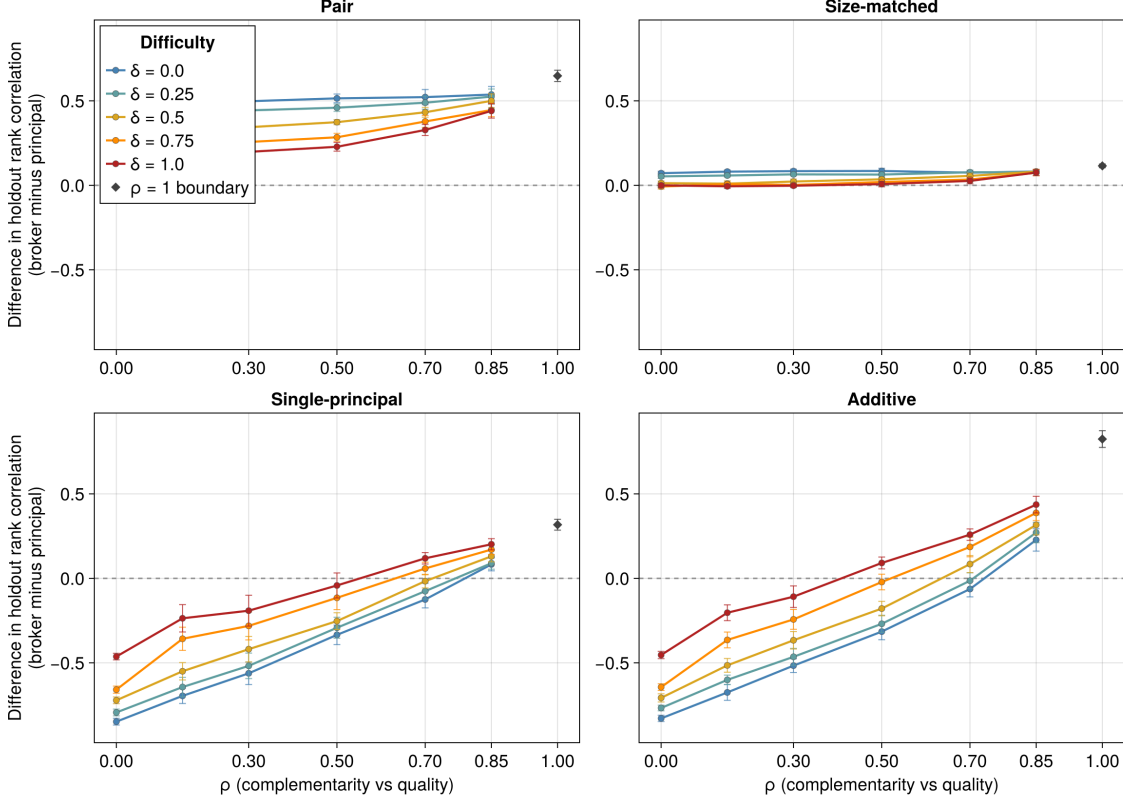


Figure RA2: Broker-minus-principal differences in holdout rank correlation across the $\rho \times \delta$ design. Each panel shows one Ridge broker variant, each line shows one value of δ over $\rho = 0.0$ through 0.85, and markers are late-window seed means. The pure-quality boundary $\rho = 1.0$, where δ is inactive, is shown once and unconnected. The vertical scale is shared across panels, whiskers are 95% Monte Carlo intervals, and the dashed line marks equal broker and principal holdout rank correlations.

Interpretation. At the baseline, every restriction reduces the broker’s ranking advantage. Size matching lowers the broker-minus-principal difference in holdout rank correlation to 0.035, a paired change of -0.338 from base Ridge $[-0.354, -0.323]$. The single-principal and additive differences are -0.253 and -0.179; their paired changes are -0.627 $[-0.680, -0.574]$ and -0.553 $[-0.600, -0.505]$. All intervals are 95% Monte Carlo intervals.

Size matching reduces the ranking difference in 31/31 effective realizations. The median change is -0.370, and the difference remains positive in 27/31 realizations. This is not only a change in broker ranking. The median broker holdout rank correlation changes by -0.082, while the median principal correlation changes by +0.289. Principal ranking also exceeds base Ridge in 31, 31, and 28 realizations under size matching, single-principal history, and additive Ridge, respectively. Altered broker rankings may expose agents to a broader range of partner types, but the saved diagnostics do not measure that range. The experiment therefore does not test this possible exploration mechanism directly.

The pair-versus-additive contrast retains both endpoint types but removes the bilinear pair features. As Figure RA2 shows, this restriction matters most when complementarity contributes strongly to match quality. The median additive-Ridge ranking difference rises from -0.707 at $\rho = 0.0$ to 0.316 at $\rho = 0.85$ and 0.825 at the pure-quality boundary $\rho = 1.0$. At that boundary, the additive

representation matches the production function and exceeds the base Ridge difference of 0.648. This could reflect the finite-sample cost of unnecessary bilinear regressors, but the system-level contrast does not isolate that mechanism.

Single-principal and additive Ridge both produce additive pair scores, but only additive Ridge retains both endpoint types in each fitting row. At the baseline, the single-principal-minus-additive ranking difference is -0.074 ([-0.107, -0.042]). Across effective realizations, the median difference is -0.053, and single-principal Ridge exceeds additive Ridge in only 1/31 realizations. Retaining both endpoint types therefore produces a modest system-level benefit among these two additive-score variants. Neither variant can represent the bilinear interaction.

The brokered-minus-self output gap remains positive in all 31 realizations under each restriction, even when the broker’s ranking difference is negative. This channel comparison is descriptive because channel choice, candidate sets, and offer acceptance are endogenous. More generally, these ablations change prediction, matching, learning histories, outsourcing, and network position together. They estimate the consequences of each restriction for the resulting system, not additive shares of a fixed-data prediction advantage.